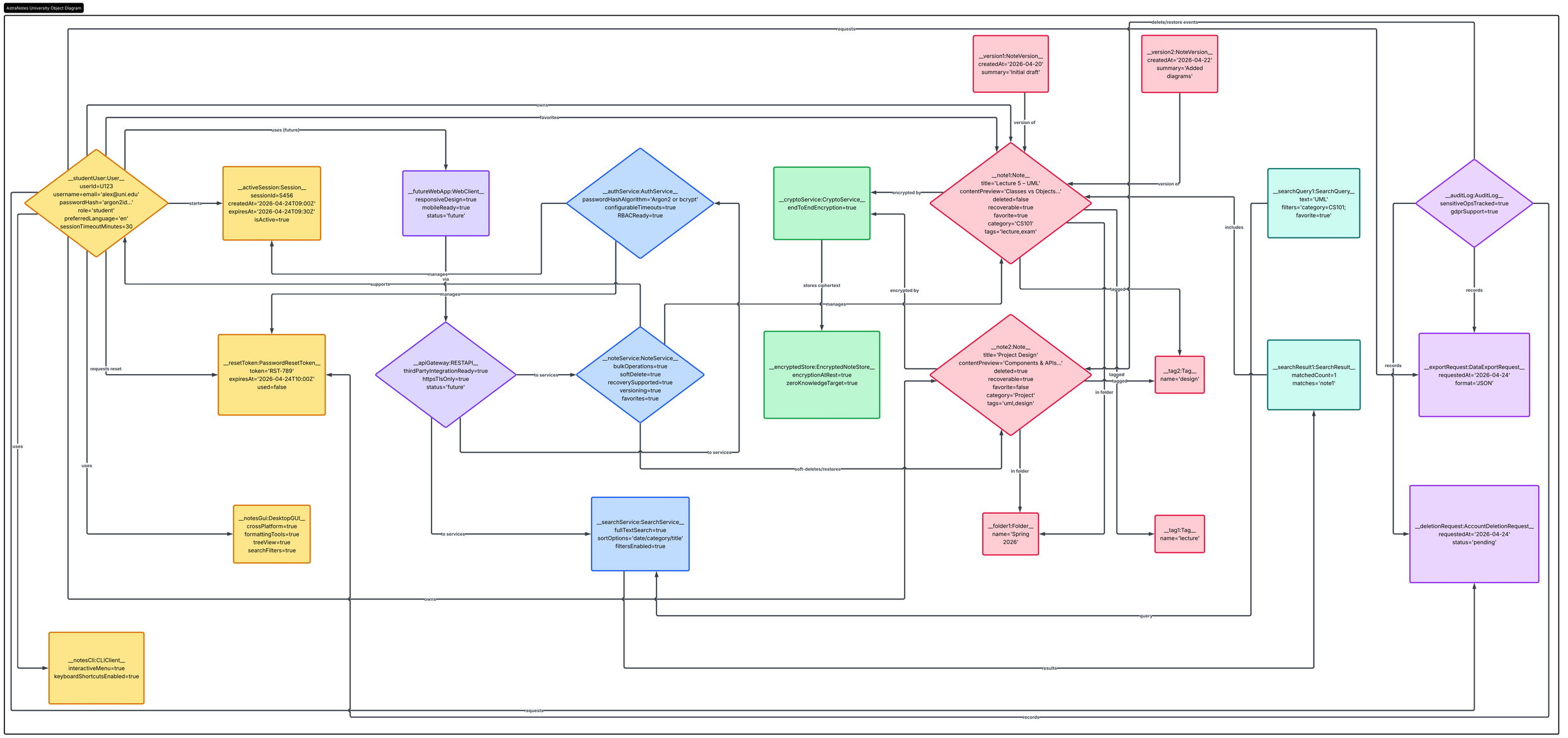
AstraNotes University Object Diagram
__studentUser:User__
userId=U123
username=email='alex@uni.edu'
passwordHash='argon2id...'
role='student'
preferredLanguage='en'
sessionTimeoutMinutes=30
__activeSession:Session__
sessionId=S456
createdAt='2026-04-24T09:00Z'
expiresAt='2026-04-24T09:30Z'
isActive=true
__futureWebApp:WebClient__
responsiveDesign=true
mobileReady=true
status='future'
__authService:AuthService__
passwordHashAlgorithm='Argon2 or bcrypt'
configurableTimeouts=true
RBACReady=true
__cryptoService:CryptoService__
endToEndEncryption=true
__note1:Note__
title='Lecture 5 – UML'
contentPreview='Classes vs Objects...'
deleted=false
recoverable=true
favorite=true
category='CS101'
tags='lecture,exam'
__version1:NoteVersion__
createdAt='2026-04-20'
summary='Initial draft'
__version2:NoteVersion__
createdAt='2026-04-22'
summary='Added diagrams'
__searchQuery1:SearchQuery__
text='UML'
filters='category=CS101; favorite=true'
__auditLog:AuditLog__
sensitiveOpsTracked=true
gdprSupport=true
__resetToken:PasswordResetToken__
token='RST-789'
expiresAt='2026-04-24T10:00Z'
used=false
__apiGateway:RESTAPI__
thirdPartyIntegrationReady=true
httpsTlsOnly=true
status='future'
__noteService:NoteService__
bulkOperations=true
softDelete=true
recoverySupported=true
versioning=true
favorites=true
__encryptedStore:EncryptedNoteStore__
encryptionAtRest=true
zeroKnowledgeTarget=true
__note2:Note__
title='Project Design'
contentPreview='Components & APIs...'
deleted=true
recoverable=true
favorite=false
category='Project'
tags='uml,design'
__tag2:Tag__
name='design'
__searchResult1:SearchResult__
matchedCount=1
matches='note1'
__exportRequest:DataExportRequest__
requestedAt='2026-04-24'
format=JSON'
__notesGui:DesktopGUI__
crossPlatform=true
formattingTools=true
treeView=true
searchFilters=true
__searchService:SearchService__
fullTextSearch=true
sortOptions='date/category/title'
filtersEnabled=true
__folder1:Folder__
name='Spring 2026'
__tag1:Tag__
name='lecture'
__deletionRequest:AccountDeletionRequest__
requestedAt='2026-04-24'
status='pending'
__notesCli:CLIClient__
interactiveMenu=true
keyboardShortcutsEnabled=true
requests
delete/restore events
owns
favorites
uses (future)
version of
starts
manages
via
supports
manages
encrypted by
stores ciphertext
manages
encrypted by
version of
includes
tagged
tagged
tagged
in folder
in folder
records
records
requests reset
to services
to services
soft-deletes/restores
uses
uses
to services
owns
query
results
requests
records